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Multi-parameter Optimization of Laser Cladding 15-5PH Using TOPSIS-GRA Based on Combined Weighting Method

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As a martensitic precipitation-hardening stainless steel of high strength, high hardness, and excellent corrosion resistance, 15-5PH stainless steel can be used in laser cladding to repair components or improve surface performance. The cladding layer is affected by the process parameters directly. A Taguchi experiment (L_{25}) of 15-5PH powder laser cladding is carried out on the surface of 45# steel to study the optimization of laser cladding process parameters. Analytic Hierarchy Process and Entropy Weighting Method are combined to determine the weight of each response index, the comprehensive Technique for Order Preference by Similarity to Ideal Solution and Grey Relational Analysis are integrated to obtain a better geometric performance of the cladding layer. The Analysis of Variance shows that the scanning speed is the most significant parameter on the performance of 15-5PH cladding layer. The validation experiments shows that under the optimal process parameters, the three geometric performance responses of the cladding layer have been improved as expected. The optimized cladding layer has obvious advantages in geometry morphology and properties, which verified the feasibility of the optimization method.

Keywords laser cladding, microstructure, optimization, properties, TOPSIS-GRA

1. Introduction

15-5PH stainless steel is a low-carbon martensitic precipitation-hardening stainless steel with high strength and moderate corrosion resistance (Ref 1). It has been used for a variety of applications including aerospace structural components, petrochemical industry components, aircraft fittings, chemical process equipment, and nuclear reactor components (Ref 2). Due to its excellent performance, 15-5PH has been gradually used in the field of metal surface modification, a large number of scholars have done a lot of research on it (Ref 3-6).

As an excellent surface coating technology, laser cladding (Ref 7, 8) has existed for more than 30 years which has been gradually applied in the industrial field since the 21st century. Compared with traditional thermal spraying technology, in addition to the flexible coating process, laser cladding offers the formation of unique coating microstructure and controlled localized heating which leads to a smaller heat affected zone. Laser cladding (Ref 9) has the following advantages: a strong metallurgical bonding force can be produced between the substrate and the cladding powder; the dilution rate of the cladding zone is controllable, as well as the heat affected zone is shallow; the laser cladding is a rapid cold and rapid heat process (Ref 10), which ensures the fine and uniform microstructure; the formation of a large number of hard phases and cementite during the laser cladding process is beneficial to obtain a coating with better mechanical properties than the substrate; what's more, laser cladding is environmentally friendly, fast and flexible (Ref 11).

In addition, as a typical surface modification technology, laser cladding can prepare high-performance coatings on low-performance and cheap base materials to reduce material costs and save precious metal materials (Ref 12). However, problems such as porosity, fracture, and surface unevenness still exist in the manufacturing process, which are largely caused by the mismatch of laser cladding process parameters (Ref 13). With the deepening of research on laser cladding technology, optimizing process parameters has become an effective mean to improve the performance of industrial products. In the laser cladding process, laser, powder, and gas phases are coupled with each other, resulting in a complex function mapping between process parameters and performance characteristics. If the accuracy of the established regression model is not high, it is difficult to be applied for further parameter optimization. Therefore, many process parameter optimization methods have been produced (Ref 14, 15), such as mathematical modeling method, response surface method, Taguchi method, machine learning, etc., which can well solve the regression problem under this complex mapping. From the perspective of empirical statistics, in order to study the relationship between laser cladding process parameters and high-performance cladding layer, Chen et al. (Ref 16) studied the effects of process parameters (laser beam power, scanning speed and powder feeding rate) on the performance of the JG-3 Iron-based powder cladding layer through interactive orthogonal experiments, and determined the best process through variance analysis. Lian et al. (Ref 17) designed a multi-track laser cladding experiment for curved surfaces using a center composite design, and analyzed the responses of different process parameters to the performance of laser cladded surface using the response surface method. Some scholars combined traditional parameter optimization methods with machine learning to achieve multi-parameter optimization of laser cladding. Peng et al. (Ref 18)

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used hybrid TS-GEP algorithm and NSGA-II to predict the result of energy consumption compared to powder utilization in laser cladding. The prediction results were compared with those predicted by the Response Surface Method (RSM). The research obtained the highest fitting performance and provided an optimal set of process parameters for maximizing energy and metal powder efficiency. Zhu et al. (Ref 19) explored the appropriate dilution rate in the laser cladding repair results, by establishing a finite element model of the laser cladding experimental results and using deep learning to obtain a more ideal dilution rate result of cladding layer. Ma et al. (Ref 20) combined RSM with PSO to prepare a high-performance, high-entropy alloy coating. Process parameter optimization through machine learning is robust and fault-tolerant, which can fit any nonlinear relationship. But the main disadvantage is that in the case of small sample experiments, the optimization accuracy is very low and the reproducibility is low. At the same time, some intelligent algorithms act as a black box system and cannot observe the intermediate processes.

In order to solve the multi-criteria evaluation problem, Hwang and Yoon proposed the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) (Ref 21). Some scholars combined TOPSIS with the above parameter optimization methods to solve optimization problems in different fields. Ananthakumar et al. (Ref 22) studied the influence of process parameters in Plasma Arc Cutting (PAC) on forming performance characteristics. According to the Box–Behnken design approach in response surface method, 30 sets of experiments were carried out, and the ideal solution method was used to determine the optimal cutting conditions. Marzban et al. (Ref 23) used the TOPSIS method of Principal Component Analysis (PCA) to optimize the process parameters of AISI 1040 laser cladding, and obtained the optimal process parameters. Since different parameters have different degrees of influence on the response results, parameters need to be weighted when using TOPSIS. Common weighting methods include factor analysis, Analytic Hierarchy Process (AHP), and Entropy Weighting Method (EWM). Yousefzadeh et al. (Ref 24) used AHP to assign weights to TOPSIS and optimized the hydrometallurgical process. Zhang et al. (Ref 25) solved the comprehensive benefit problem in the remanufacturing system by using TOPSIS based on EWM. But the shortcoming of AHP is that it is too subjective and the judgment is mainly dependent on experts, which may result in inaccurate weighting results. As a weighting method with objective advantages, EWM has the disadvantage that it cannot reflect the degree of importance the decision makers attach to different indicators, and there will be a certain weight and the opposite degree of the actual indicator. For this, when assigning weights to indicators, the inherent statistical laws and authoritative values between indicator data should be considered.

In order to solve the problem that process parameters are difficult to be determined in practical engineering application. The single-track laser cladding of 15-5PH alloy powder on the surface of 45 steel is taken as a case. Laser beam power (P), scanning speed (V), and powder feeding speed (F) are selected as input parameters to evaluate output performance such as cladding width (W), cladding depth (h), and dilution rate (η), Taguchi method is used to scientifically design experiments. A new weighting method Integrated Weighting Method (IWM) for laser cladding is proposed. The method of integrating TOPSIS and Grey Relational Analysis (GRA) is adopted to obtain the final decision index for multi-parameter optimization. Finally, the microstructure and properties of the optimized laser cladding are analyzed.

2. Multi-parameter Optimization of Laser Cladding Using TOPSIS-GRA Based on AHP and EWM

2.1 Problem Definition

In the laser cladding process, the coupling of multiple process parameters (laser parameters, cladding process parameters, material parameters) affects the cladding results. In order to achieve the goal of maximizing each goal as much as possible and seek the optimal compromise solution for multiple goals, the laser cladding multi-objective optimization mathematical model as shown in Eq 1, is constructed.

$$\begin{aligned} &\max S = f(Q) = (f_1(Q), f_2(Q), \ldots f_k(Q)) \\ \text{s.t.}\quad &Q = (Q_1, Q_2, \ldots, Q_i, \ldots, Q_n) \in Q \\ &S = (S_1, S_2, \ldots, S_j, \ldots, S_m) \in S \end{aligned} \qquad \text{(Eq 1)}$$

where Q represents the variable that is the ith process parameter, S refers to the target vector to the jth cladding performance target, f is the functional relationship between the various process parameters of laser cladding and the performance characteristics of the cladding layer.

The process parameters of laser cladding such as laser generator parameters, cladding process parameters, and material parameters affect the forming performance and macroscopic morphology of the cladding layer. Once the laser cladding equipment, substrate material, and alloy powder materials are determined, the characteristics of the entire laser cladding system are determined and there are few adjustable parameters. In laser cladding process, the process parameters such as laser beam power, scanning speed, powder feeding speed, laser spot diameter, the distance of the head from the machined surface, carrier gas rate and so on are the key to control the forming quality of cladding layer. Three process parameters of laser beam power (P), scanning speed (V), and powder feeding speed (F) which are frequently adjusted in engineering applications are selected as input arguments in this study (Ref 16, 17).

The macro morphology of the cladding layer including geometric and appearance features is an important feature indicator to describe the performance of the laser cladding layer, which is closely related to the cross-sectional geometric characteristics. Fig. 1 shows a typical cross section of the cladding layer, W is cladding width, h is cladding depth, and η is dilution rate. Larger W and lower h can ensure that laser cladding reduces defects in the multi-track cladding. Too large η will lead to a waste of powder material, too small η will lead to insufficient metallurgical bond strength between two media. In order to ensure that the multi-track cladding surface is flat, we choose 30% (Ref 26, 27) as the target value of η. The multi-objective optimization model of laser cladding is shown in Eq 2.

$$\begin{cases} \max f_W(P, V, F) \\ \min f_h(P, V, F) \\ \min \left| f_\eta(P, V, F) - 30\% \right| \end{cases} \qquad \text{(Eq 2)}$$

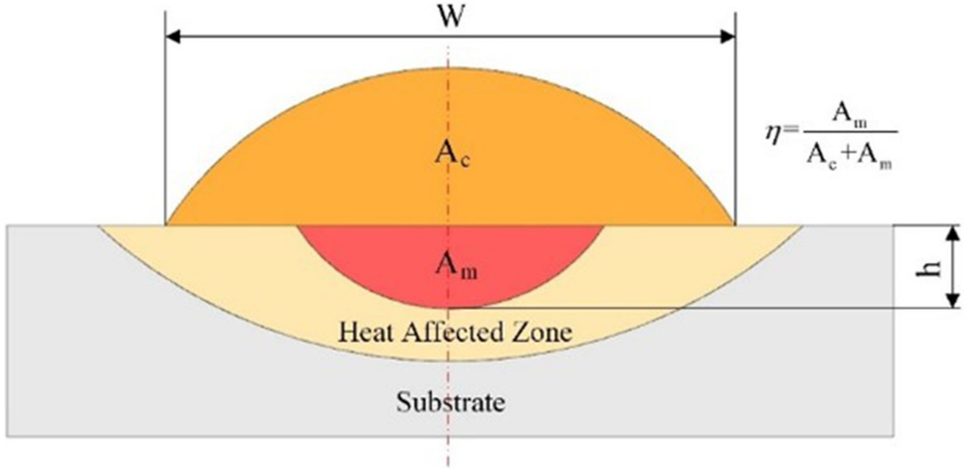


Fig. 1. Typical geometry characteristics of single-track cladding cross-section

2.2 Multi-objective Optimization Procedure of Laser Cladding

According to the multi-objective optimization problem constructed by Eq 2, AHP and EWM are integrated to assign weights to the optimization objectives. A multi-objective optimization procedure for laser cladding is proposed, as shown in Fig. 2, which consists of the following four steps.

- Step 1: Establishment of multi-parameter optimization model for laser cladding

For laser cladding, the complex relationship between its process parameters and cladding performance characteristics is analyzed. The process parameters of laser cladding are taken as input and the geometric characteristics of the cladding layer are taken as output. Constructing a unique multi-parameter optimization model of laser cladding is the first step to complete the optimization problem.

- Step 2: Weight determination and adjustment of performance characteristics in laser cladding results

The performance characteristics of the laser cladding results will represent various forms, so it is necessary to analyze the importance of each optimization target and assign a weight to each optimization object. Here, a new Integrated Weighting Method (IWM) that combines AHP and EWM is proposed.

- Step 3: Determination of the final optimization results for laser cladding

In order to obtain the process parameters for the best forming performance results of single-track laser cladding, and at the same time provide process parameter references for multi-track lap laser cladding, the final optimization results of process parameters will be given.

By combining TOPSIS and GRA, the Euclidean distance closeness index and situational closeness index between experimental parameters and ideal solution are calculated respectively, and then a comprehensive closeness index is synthesized. The optimization of laser cladding process parameters is calculated by the ranking of comprehensive closeness index.

- Step 4: Validation experiment under optimized process parameters

According to the final process parameter optimization results of Step 3, verification experiments are needed to determine the feasibility of the process plan and the multi-parameter optimization method of laser cladding.

2.2.1 Calculation of combined weight value based on AHP & EWM. Both AHP and EWM are scientific methods for weighting the performance characteristics of laser cladding. The difference is that the former is subjective and the latter is objective. In order to analyze the different importance of cladding performance features more accurately, a new Integrated Weighting Method (IWM) that combines AHP and EWM is proposed, which combines both advantages of AHP and EWM to calculate the different weights of cladding performance features.

- Step 1: Calculation of weight value based on AHP

Since the importance judgments between the geometric performance characteristics of each cladding layer is entirely dependent on the experience of the experts, so the selection of the experts is deemed to be of utmost importance. Here, the expert group consists of ten engineers with more than five years of laser cladding remanufacturing industry. In order to arrive at consensual weights based on the individual assessments obtained from the ten experts, one mechanism is used, whereby the geometrical mean of all ten entries (i.e., the immediate results of pairwise comparisons) for each cell of the various comparison matrices is computed and a new, ‘‘mean’’ comparison matrix is obtained for each set of factors.

For explanation, the steps of determining the weights for individual index with the AHP methodology can be shown as the following example.

Firstly, the prioritization matrix can be set up, as shown in Table 1.

Secondly, the index priority comparison matrix U as shown in Table 2 can be established after the pairwise comparisons between the m elements.

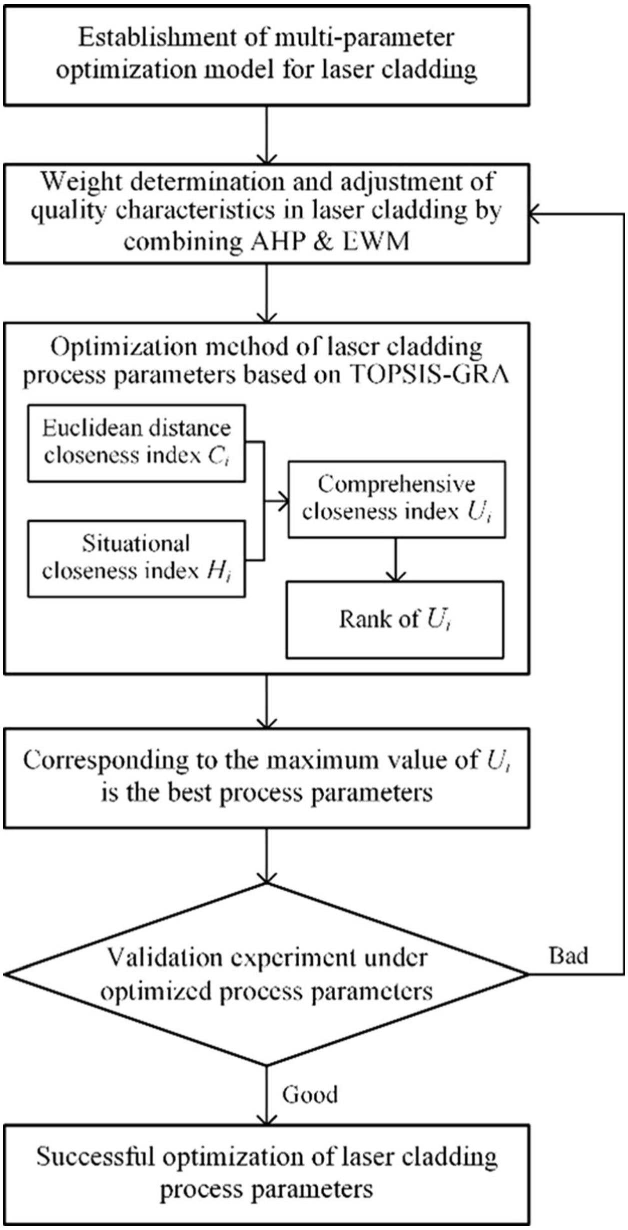


Fig. 2. Multi-objective optimization procedure of laser cladding

Then, based on AHP, the weight value ω_j of cladding layer geometric performance characteristics can be calculated by Eq 3.

$$\omega_j = \overline{W_j} / \sum_{j=1}^{m} \overline{W_j} \quad j = 1, 2, \ldots, m$$
$$\text{where } \overline{W_j} = \sqrt[m]{M_j}, M_j = \prod_{j=1}^{m} u_j \tag{Eq 3}$$

- Step 2: Calculation of weight value based on EWM

Entropy Weighting Method (EWM) is an objective weighting method. This method uses the information entropy value E_j to measure the amount of information, and then determines the importance of evaluation indicators. Then calculate the weight value θ_j of each performance feature index through information entropy.

$$\theta_j = \frac{1 - E_j}{n - \sum E_j} \; (j = 1, 2, \ldots, m) \tag{Eq 4}$$

where $E_j = -\frac{1}{\ln n} \sum_{i=1}^{n} p_{ij} \ln(p_{ij}) (j = 1, 2, \cdots, m)$, $p_{ij} = y_{ij} / \sum_{i=1}^{n} y_{ij}$, the definition of y_{ij} is in step1 in section 2.2.2.

- Step 3: Calculation of weight value w_j based on IWM

The integrated weight value w_j of each cladding layer geometric performance characteristic index can be calculated by Eq 5.

$$w_j = \frac{\omega_j \theta_j}{\sum_{j=1}^{m} \omega_j \theta_j} \tag{Eq 5}$$

Table 2. The priority comparison index of laser cladding performance feature

u_{ij}	W	h	η
W	1	3	1/7
h	1/3	1	1/9
η	7	9	1

Table 1. The scale of prioritization comparison (Ref 28, 29)

Intensity of importance	Definition	Explanation
1	Equal importance	u_i and u_j contribute equally to the objective
3	Moderate importance	u_i is slightly favored over u_j
5	Strong importance	Experience and judgment strongly favor u_i over u_j
7	Very strong or demonstrated importance	u_i is favored very strongly over u_j
9	Extreme importance	The evidence favoring u_i over u_j is of the highest possible order of affirmation
2,4,6,8	The median of the above two adjacent judgments	
Reciprocal	The comparison between the factor u_i and u_j is u_{ij}, and the comparison between the factor u_j and u_i is $u_{ji} = 1 / u_{ij}$	

Fig. 3. Laser cladding experiment platform

2.2.2 Calculation of Euclidean distance closeness index based on TOPSIS.

- Step 1: Construction of the original decision matrix

Different decision responses correspond to different evaluation indicators. The laser cladding process parameter Q=(Q_1,…, Q_i,…, Q_n) is set, and the cladding performance characteristic S=(S_1,…, S_j,…, S_m) is set. The value of the decision response set S_j corresponding to the decision parameter set Q_i is y_{ij} (i=1,2,…,n; j=1,2,…,m). Then the laser cladding original decision matrix Y is:\fleqno$\rm(Eq{\hbox { }}5)

Step 2: Construction of the orthogonalized matrix

The characteristics that respond to expectations are divided into Larger-the-Better Characteristic (LBC), Smaller-the-Better Characteristic (SBC) and Nominal-the-Type Characteristic (NTC). This different direction makes the analysis confusing. In order to simplify the analysis, we normalize the original data and unify the larger-the-better characteristic. According to Eq 7, the orthogonalized matrix X is as follows:

$$X = \left[x_{ij}\right]_{n\times m} = \begin{cases} \max(y_j) - y_{ij} & \text{SBC} \\ \frac{1-\left|y_{ij}-y_{\text{best}}\right|}{\max\left(\left|y_{ij}-y_{\text{best}}\right|\right)} & \text{NTC} \end{cases} \qquad \text{(Eq 7)}$$

where y_{best} is the desired value in nominal-the-type characteristic of dilution rate (η), which equals to 30%.

- Step 3: Calculation of the weighted normalized matrix

Due to the difference in the physical meaning between the reference sequence and the comparison sequence, there is a certain gap in the dimension, which leads to the annihilation of some factors and the inaccuracy of the analysis results. Therefore, it is necessary to control the signal noise, according to the different output response characteristics (Ref 18). The X matrix is normalized.

$$N = [n_{ij}]_{n\times m} = \frac{x_{ij} - \min x_{ij}}{\max_i x_{ij} - \min_i x_{ij}} \qquad \text{(Eq 8)}$$

The new weighted decision matrix A is calculated by Eq 9.

$$A = [a_{ij}]_{n\times m} = w_j \times N \qquad \text{(Eq 9)}$$

where w_j is the combined weight value of S_j calculated by Eq 5.

Step 4: Construct positive ideal solution set A^+ and negative ideal solution set A^-

$$\begin{aligned} A^+ &= [a_1^+, a_2^+, ..., a_m^+] \\ A^- &= [a_1^-, a_2^-, ..., a_m^-] \end{aligned} \qquad \text{(Eq 10)}$$

Due to larger-the-better characteristic, $a_j^+ = \max\limits_{1\le j\le m}\{S_j\}$ and $a_j^- = \min\limits_{1\le j\le m}\{S_j\}$.

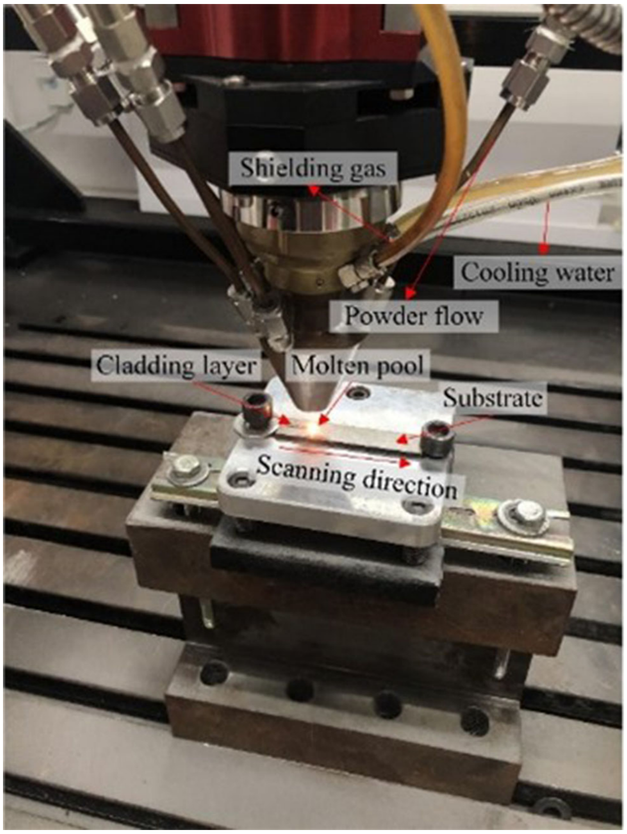


Fig. 4. Single-track laser cladding process

Table 3. Elemental composition of the substrate material and cladding powder

	Elements wt. ,%							
Material	**C**	**Mn**	**Si**	**Cr**	**Ni**	**Cu**	**Nb**	**Fe**
45 Steel	0.46	0.65	0.27	0.17	…	…	…	Bal.
15-5PH	0.026	0.84	0.38	15.41	4.66	3.71	0.30	Bal.

Step 5: Calculate the Euclidean distance from the ith scheme in the weighted decision matrix to the positive ideal solution and the negative ideal solution

Distance to the positive ideal solution is calculated by Eq 11.

$$d_i^+ = \sqrt{\sum_{j=1}^{n} \left(a_{ij} - a_j^+\right)^2}, \quad i = 1, 2, \ldots, n \qquad \text{(Eq 11)}$$

Distance to the negative ideal solution is calculated by Eq 12.

$$d_i^- = \sqrt{\sum_{j=1}^{n} \left(a_{ij} - a_j^-\right)^2}, \quad i = 1, 2, \ldots, n \qquad \text{(Eq 12)}$$

Step 6: Calculation of Euclidean distance closeness index

Euclidean distance closeness index C_i between the ith group of experimental parameters and the ideal solution set can be defined, as shown in Eq 12. The larger the index, the greater the contribution of this group of parameters to the geometric performance of single-track laser cladding.

$$C_i = \frac{d_i^-}{d_i^+ + d_i^-} \qquad \text{(Eq 13)}$$

2.2.3 Calculation of comprehensive closeness index based on TOPSIS-GRA method. The single TOPSIS (Ref 21) method can only assess the degree of closeness between the object and the optimal solution set in Euclidean distance. The Grey Relational Grade (GRG) can well reflect the closeness of the development situation of the two. For this, the TOPSIS-GRA method is adopted to define the comprehensive closeness index of the designed parameters (Ref 22).

Step 1: Calculation of grey relational coefficient matrix based on ideal solutions

In order to obtain the correlation between the experimental object and the ideal solution, the calculation of the grey relational coefficient needs to set the reference sequence and the comparison sequence, and calculate the deviation sequence. Therefore, the positive ideal solution set A^+ and the negative ideal solution set A^- are used as the reference sequence, and the value of the decision response set S_j corresponding to the decision parameter set Q_i in the decision matrix A is used as the comparison sequence. The grey relational coefficient matrices G^+ and G^- relative to the positive ideal solution set and the negative ideal solution set are calculated.

$$\begin{cases} G^+ = (g_{ij}^+)_{n\times m} \\ G^- = (g_{ij}^-)_{n\times m} \end{cases} \qquad \text{(Eq 14)}$$

where

$$g_{ij}^+ = \frac{\min\limits_i \min\limits_j \left|a_j^+ - a_{ij}\right| + \rho \max\limits_i \max\limits_j \left|a_j^+ - a_{ij}\right|}{\left|a_j^+ - a_{ij}\right| + \rho \max\limits_i \max\limits_j \left|a_j^+ - a_{ij}\right|}$$
$$g_{ij}^- = \frac{\min\limits_i \min\limits_j \left|a_j^- - a_{ij}\right| + \rho \max\limits_i \max\limits_j \left|a_j^- - a_{ij}\right|}{\left|a_j^- - a_{ij}\right| + \rho \max\limits_i \max\limits_j \left|a_j^- - a_{ij}\right|} \qquad \text{(Eq 15)}$$

where g_{ij}^+ and g_{ij}^- are the grey relational coefficients of the distance between each experimental object and the positive and negative ideal solution, ρ is the distinguishing coefficient, which is defined in the range $0 \le \rho \le 1$, ρ is taken as 0.5 in these equations.

Step 2: Calculation of GRG between the evaluation object and the ideal solution

$$\begin{cases} \delta_i^+ = \dfrac{1}{m}\sum\limits_{j=1}^{m} g_{ij}^+ \\ \delta_i^- = \dfrac{1}{m}\sum\limits_{j=1}^{m} g_{ij}^- \end{cases} \qquad \text{(Eq 16)}$$

where δ_i^+ and δ_i^- are the grey relational grades of the positive ideal solution and the negative ideal solution.

Step 3: Calculation of situational closeness index based on GRG

Equation (17) can be used to define the situational closeness index H_i of the ith group of experimental parameters to the ideal solution set.

Table 4. Factors and levels

Factors	Notations	Unit	Level 1	2	3	4	5
Laser beam power	P	W	450	500	550	600	650
Scanning speed	V	mm/min	500	550	600	650	700
Powder feeding rate	F	r/min	1	1.1	1.2	1.3	1.4

Table 5. Design of laser cladding experiment $L_{25}(5^3)$, i.e. Taguchi orthogonal array

No.	P, W	V, mm/min	F, r/min	NO.	P, W	V, mm/min	F, r/min
1	450	500	1	14	550	650	1
2	450	550	1.1	15	550	700	1.1
3	450	600	1.2	16	600	500	1.3
4	450	650	1.3	17	600	550	1.4
5	450	700	1.4	18	600	600	1
6	500	500	1.1	19	600	650	1.1
7	500	550	1.2	20	600	700	1.2
8	500	600	1.3	21	650	500	1.4
9	500	650	1.4	22	650	550	1
10	500	700	1	23	650	600	1.1
11	550	500	1.2	24	650	650	1.2
12	550	550	1.3	25	650	700	1.3
13	550	600	1.4				

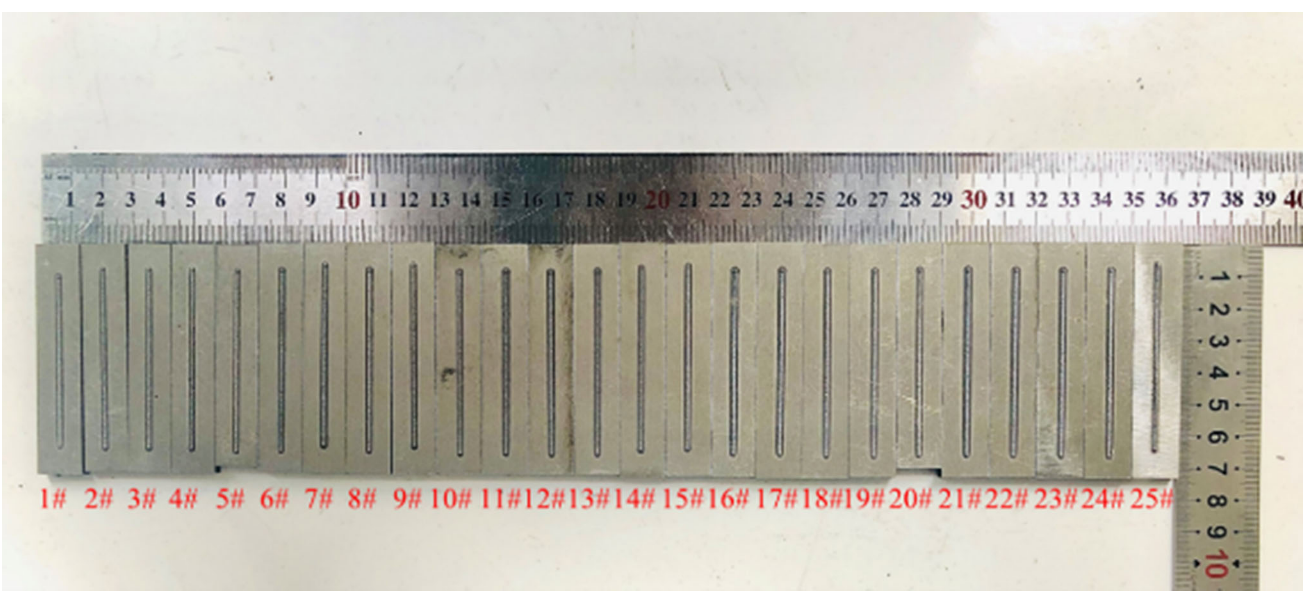


Fig. 5. Macro results of the experiment

$$H_i = \frac{\delta_i^+}{\delta_i^+ + \delta_i^-} \quad \text{(Eq 17)}$$

Step 4: Comprehensive closeness index calculation based on Euclidean distance and situation closeness index

To define the comprehensive closeness index U_i between the ith group of experimental parameters and the ideal solution set, Eq 18 can be usde. The larger the index, the closer the set of parameters to the optimal response.

$$U_i = \alpha C_i + (1 - \alpha) H_i \quad \text{(Eq 18)}$$

where the value of weight α is 0.5.

Table 6. Raw data of cladding cross-section

No.	W, μm	h, μm	A_c, μm^2	A_m, μm^2	η
1	1080.503	44.720	169153.063	32449.720	16.096%
2	1020.780	34.951	145218.557	25420.334	14.897%
3	979.151	19.551	190024.924	12962.606	6.386%
4	1063.280	45.752	162483.109	26597.398	14.067%
5	952.961	29.604	167966.315	17670.322	9.519%
6	1082.545	203.607	212170.521	88115.901	29.344%
7	1095.563	162.646	220651.745	86883.875	28.252%
8	1056.248	138.505	186911.030	69178.639	27.013%
9	1042.166	153.747	169467.400	67921.477	28.612%
10	1011.702	174.082	100213.670	91822.900	47.815%
11	1079.013	198.223	285257.124	107870.490	27.439%
12	1122.206	194.407	261790.904	105713.856	28.765%
13	1141.269	196.949	265590.651	87349.760	24.749%
14	1123.477	217.279	135217.199	119586.241	46.933%
15	1091.707	222.365	123180.901	108402.584	46.809%
16	1159.079	229.989	300699.953	123767.900	29.158%
17	1184.564	284.633	272036.331	157968.040	36.736%
18	1185.758	280.856	199195.208	155727.433	43.876%
19	1167.960	259.209	164962.771	143775.914	46.569%
20	1133.833	297.019	151810.693	150040.723	49.707%
21	1236.601	321.493	335182.692	190324.019	36.217%
22	1256.952	339.319	221703.821	205662.690	48.123%
23	1207.356	297.332	248939.913	164577.629	39.799%
24	1180.710	298.643	239851.527	165232.451	40.790%
25	1098.065	296.061	183756.417	164882.836	47.293%

Table 7. Combined weight value based on AHP&EWM

Response indicator	Weighting method		
	AHP	EWM	IWM
Cladding width , W	0.1488	0.0096	0.0043
Cladding depth , h	0.0658	0.6205	0.1227
Dilution rate , η	0.7854	0.3699	0.8730

3. Experiment Design of Laser Cladding

3.1 Experiment Condition

The laser cladding device for experiment is iLam® 509C series equipment, which consists of a three-axis (expandable fourth and fifth axis) movement mechanism built by electric cylinder, self-developed automatic powder feeder, laser head, nozzle, motion control system, water cooler, laser and other components. The rated power of RFL-A1000D fiber laser is 1000 W, the wavelength is 915 ± 10 nm. The laser cladding experiment platform is shown in Fig. 3 and the single-track laser cladding process is shown in Fig. 4.

The cooling system is turned on in the whole process to prevent overheating and damage of the laser head during the process, and argon gas is introduced as the shielding gas to avoid rapid oxidation of the cladding area. The laser beam is used as the heat source from the laser generator. The 45# steel substrate surface is irradiated by the laser beam to form a molten pool, at the same time, the 15-5PH powder carried by argon gas is melted. The laser head and the workbench move relative by the control system. As the laser beam moves, the cladding layer is formed with the molten pool rapidly solidifies by contacting the air.

3.2 Material and Procedure

The experiment substrate is made of high-performance carbon structure 45 steel, with a size of $75\times15\times5mm^3$. The cladding powder is selected from Fe-based self-fusing powder 15-5PH. Its shape is close to spherical and the size is 120-300 screen mesh. The elemental composition of the substrate material and cladding powder is shown in the Table 3.

Before the experiment, the surface of the substrate to be cladding is polished with a grinder and wiped clean with absolute ethanol to remove the surface impurities. The 15-5PH powder is put into the drying oven for drying and standby, the drying temperature is set at 150 °C, the time is 2 h, to ensure that the powder is completely dry and moisture-free to reduce defects such as pores and cracks in the cladding layer. Through the integrated CAAM programming software in the laser cladding device, the single-track cladding experiment is carried out on the substrate. The track length is 65 mm. After cladding, the substrate is cut into $15 \times 15 \times 5$ mm^3 sample pieces using wire-cutting EDM, and the cutting section is sanded from coarse to fine using 400 to 1200 mesh sandpapers, then polished by diamond abrasive paste. After etched with 4% ethanol nitrate solution, the geometry is observed and measured by the Leica DVM6S digital microscope.

Three process parameters of the laser cladding experiment are selected, namely, laser beam power (P), scanning speed (V) and powder feeding speed (F) as the influencing factors, and the experiment values of three process parameters are represented in Table 4 at five levels. L_{25} (5^3) field orthogonal experiments are designed by statistical software MINITAB 18, and the experiment parameter portfolios are shown in Table 5.

4. Results and Discussion

4.1 Experimental Results of L_{25} Single-Track Laser Cladding

25 groups of experimental results are shown in Fig. 5. The cladding cross section is measured by the Leica DVM6S digital microscope, respectively, and the geometric parameter results after measurement calculations are shown in Table 6.

4.2 Results Analysis Using TOPSIS-GRA Based on Combined Weighting Method

According to AHP and EWM in section 2.2.1, the weights of the three response indicators W, h, and η are respectively calculated, and the final weight values are calculated in combination with the Integrated Weighting Method (IWM). The weights determined thereby are shown in Table 7.

The normalized response indicators after weighted is obtained in the 2-4 columns of Table 8. Ideal solution sets are constructed according to section 2.2.2 Step 4, Euclidean distance is calculated between each experimental sample and ideal solution sets, which is shown in the 5-6 columns of Table 8. According to section 2.2.3 Step 1-2, the grey relational grade between each experimental sample and each ideal solution is shown in the 7-8 columns of Table 8.

Euclidean distance closeness index and situation closeness index, as well as the comprehensive closeness index calculated

Table 8. Calculation of normalized and weighted responses, Euclidean distances and grey relational grades

	Normalized and weighted response value			Euclidean distance		Grey relational grade	
No.	W	h	η	d_i^+	d_i^-	δ_i^+	δ_i^-
1	0.0018	0.1130	0.3692	0.5039	0.3862	0.8123	0.7773
2	0.0010	0.1168	0.3237	0.5494	0.3441	0.8073	0.7870
3	0.0004	0.1227	0.0000	0.8730	0.1227	0.7748	0.9266
4	0.0016	0.1126	0.2921	0.5810	0.3130	0.8001	0.7968
5	0.0000	0.1188	0.1191	0.7539	0.1683	0.7827	0.8572
6	0.0018	0.0521	0.8730	0.0707	0.8746	0.9517	0.7409
7	0.0020	0.0678	0.8315	0.0689	0.8342	0.9321	0.7351
8	0.0015	0.0771	0.7844	0.0997	0.7882	0.9100	0.7347
9	0.0013	0.0712	0.8452	0.0586	0.8482	0.9425	0.7325
10	0.0008	0.0634	0.2205	0.6552	0.2294	0.7578	0.8452
11	0.0018	0.0541	0.8006	0.0998	0.8024	0.9054	0.7461
12	0.0024	0.0556	0.8510	0.0706	0.8528	0.9381	0.7402
13	0.0027	0.0546	0.6983	0.1875	0.7004	0.8585	0.7558
14	0.0024	0.0468	0.2541	0.6236	0.2584	0.7537	0.8432
15	0.0020	0.0449	0.2588	0.6192	0.2626	0.7529	0.8434
16	0.0029	0.0419	0.8660	0.0811	0.8670	0.9416	0.7469
17	0.0033	0.0210	0.6418	0.2526	0.6422	0.8208	0.7838
18	0.0033	0.0224	0.3703	0.5126	0.3710	0.7586	0.8282
19	0.0030	0.0307	0.2679	0.6121	0.2697	0.7474	0.8490
20	0.0026	0.0162	0.1486	0.7322	0.1495	0.7253	0.9015
21	0.0040	0.0068	0.6615	0.2411	0.6616	0.8211	0.7910
22	0.0043	0.0000	0.2088	0.6755	0.2088	0.7257	0.8889
23	0.0036	0.0161	0.5253	0.3637	0.5256	0.7863	0.8034
24	0.0032	0.0156	0.4877	0.3999	0.4879	0.7772	0.8102
25	0.0021	0.0166	0.2404	0.6415	0.2409	0.7359	0.8679

Table 9. Results of Euclidean distance, situation and comprehensive closeness index

No.	C	H	U	Rank	No.	C	H	U	Rank
1	0.4339	0.5110	0.4724	13	14	0.2929	0.4720	0.3825	19
2	0.3851	0.5064	0.4457	15	15	0.2978	0.4717	0.3848	18
3	0.1232	0.4554	0.2893	25	16	0.9145	0.5576	0.7361	5
4	0.3501	0.5010	0.4256	16	17	0.7177	0.5115	0.6146	10
5	0.1825	0.4773	0.3299	23	18	0.4198	0.4780	0.4489	14
6	0.9252	0.5623	0.7438	2	19	0.3059	0.4682	0.3870	17
7	0.9237	0.5591	0.7414	3	20	0.1695	0.4459	0.3077	24
8	0.8877	0.5533	0.7205	6	21	0.7329	0.5093	0.6211	9
9	0.9354	0.5627	0.7490	1	22	0.2362	0.4495	0.3428	22
10	0.2594	0.4727	0.3660	20	23	0.5910	0.4946	0.5428	11
11	0.8894	0.5482	0.7188	7	24	0.5495	0.4896	0.5196	12
12	0.9235	0.5590	0.7412	4	25	0.2730	0.4588	0.3659	21
13	0.7888	0.5318	0.6603	8					

according to Eq 18 are shown in Table 9, and then the comprehensive closeness index is ranked. The results clearly show that the experiment serial number corresponding to the highest value of 0.7490 for comprehensive closeness index is No.9, whose process parameter level is $P2V4F5$, and the specific process parameters are as follows: laser beam power = 500W, scanning speed = 650 mm/min, and powder feeding rate = 1.4 r/min.

The comparison of optimal process parameter by TOPSIS-GRA-AHP and TOPSIS-GRA-EWM and TOPSIS-GRA-IWM is shown in Table 10. It can be seen from the cross section of the cladding layer that under the optimal parameter obtained by AHP, the cladding layer has too deep molten pool, while under the optimal parameter obtained by EWM, there is almost no molten pool. The cladding layer that under the optimal parameter obtained by IWM has a better result observationally.

Analysis of Variance (ANOVA) is performed for comprehensive closeness index, and the results are shown in Table 11. According to the (lower) p-quantile table of F-distribution. The numerator degree of freedom of the F-distribution is the number of independent variables, which equals to 3, and the denominator degree of freedom of the F-distribution is the horizontal level of independent variables, which equals to 5. $F_{0.025}(3,5) = 7.76$, $F_{0.01}(3,5) = 12.01$, where $F(F) < F_{0.025}(3,5) < F(P) < F_{0.01}(3,5) < F(V)$. It shows that among the three process parameters of laser beam power (P), scanning speed (V), and

Table 10. Comparison of optimal process parameter by TOPSIS-GRA-AHP & TOPSIS-GRA-EWM & TOPSIS-GRA-IWM

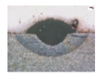
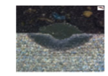
	Optimization method		
	TOPSIS-GRA-AHP	**TOPSIS-GRA-EWM**	**TOPSIS-GRA-IWM**
Optimal process parameter	*P4V1F4*	*P1V1F1*	*P2V4F5*
Cladding width (*W*)	1159.079	1080.503	1042.166
Cladding depth (*h*)	229.989	44.720	153.747
Dilution rate (η)	29.158%	16.096%	28.612%
Cladding cross-section			

Table 11. ANOVA for comprehensive closeness index

Source	DF	Seq SS	Adj MS	F	P		Contribution (%)
P	4	0.212	0.053	11.59	0	Significant	35.27
V	4	0.260	0.065	14.17	0	Significant	43.26
F	4	0.129	0.032	7.06	0.004	Significant	21.47
Error	12	0.055	0.005				
Total	24	0.656					

Table 12. Optimization results validation

Comparative items Setting level	Initial *P2V5F1*	Optimal *P2V4F5*	Improvement (%)
W(μm)	1011.702	1057.624	4.539
h(μm)	174.082	146.812	15.665
η	47.815%	29.743%	37.796

powder feeding rate(*F*), laser beam power and scanning speed play key roles in the acquirement of the best geometric performance characteristics. Among them, scanning speed is the most influential, while the effect of powder feeding rate is not obvious. The reason is that in the cladding process, the utilization rate of the blown powder is not 100%. After the protective gas is blown out, a part of the powder hits the substrate and ejects and does not participate in the cladding. The contribution of *P*, *V*, and *F* to the geometric performance characteristics of the cladding layer are 35.27%, 43.26% and 21.47%, respectively.

5. Verification Experiment of the Optimal Process Parameters

5.1 Comparison of Macroscopic Morphology Results

The combination of laser cladding process parameters recommended by the engineer is *P2V5F1* (*P*: 500 W, *V*: 700 mm/min, *F*: 1.0 r/min). According to the optimization method, the optimal process parameter is *P2V4F5* (*P*: 500 W, *V*: 650 mm/min, *F*: 1.4 r/min), so it is also necessary to conduct a verification comparison test on this process parameter. A comparison table of the validation test results is shown in Table 12. The three output indicators in the optimization test were improved. Among them, the cladding width (*W*) increased from 1011.702 μm to 1057.624 μm, with a slight increase of 4.539%; the cladding depth (*h*) decreased from 174.082 μm to 146.812 μm, an obvious improvement of 15.665%; the dilution rate (η) decreased from 47.815 to 29.743%, which closes to the standard dilution rate of 30%, with an improvement of 37.796%. The morphology of single-track laser cladding under the combination of two process parameters is shown in Fig. 6.

5.2 Microstructure Analysis

As shown in Fig.7(a), under the laser cladding test with the optimal process parameters, the entire cladding layer and the substrate formed a relatively complete metallurgical bond, without pores, slagging and other defects. In laser cladding, the influence of temperature gradient (*G*) and solidification rate (*R*) determines the morphology and size of the crystalline tissue (Ref 30, 31). From the bottom to the top of the cladding layer, there are three stages.

1. As shown in Fig. 7(b), the solidification process first starts at the bonding zone between the substrate and the cladding layer, when the temperature gradient (*G*) is extremely high and the solidification rate (*R*) is relatively low (it means the crystallization parameter *G*/*R* is very large), there is almost no overcooling of the components, and the crystal nucleation speed is much faster than the growth rate (Ref 32). Then the crystals grow at the interface with plane crystals towards solidification., a large number of oversaturated elements is precipitated along the plane crystal growth direction in the form of strip alloy carbides.

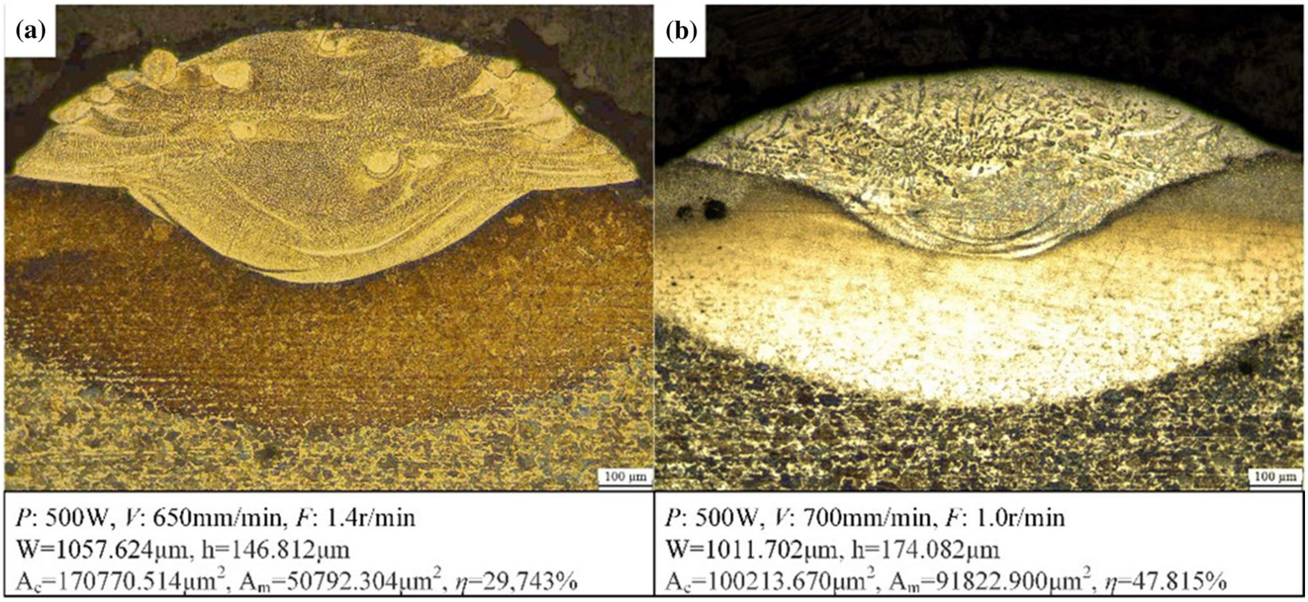


Fig. 6. Macro morphology of cladding layer cross section. (a) The optimal process parameter, (b) the process parameters recommended by the engineer

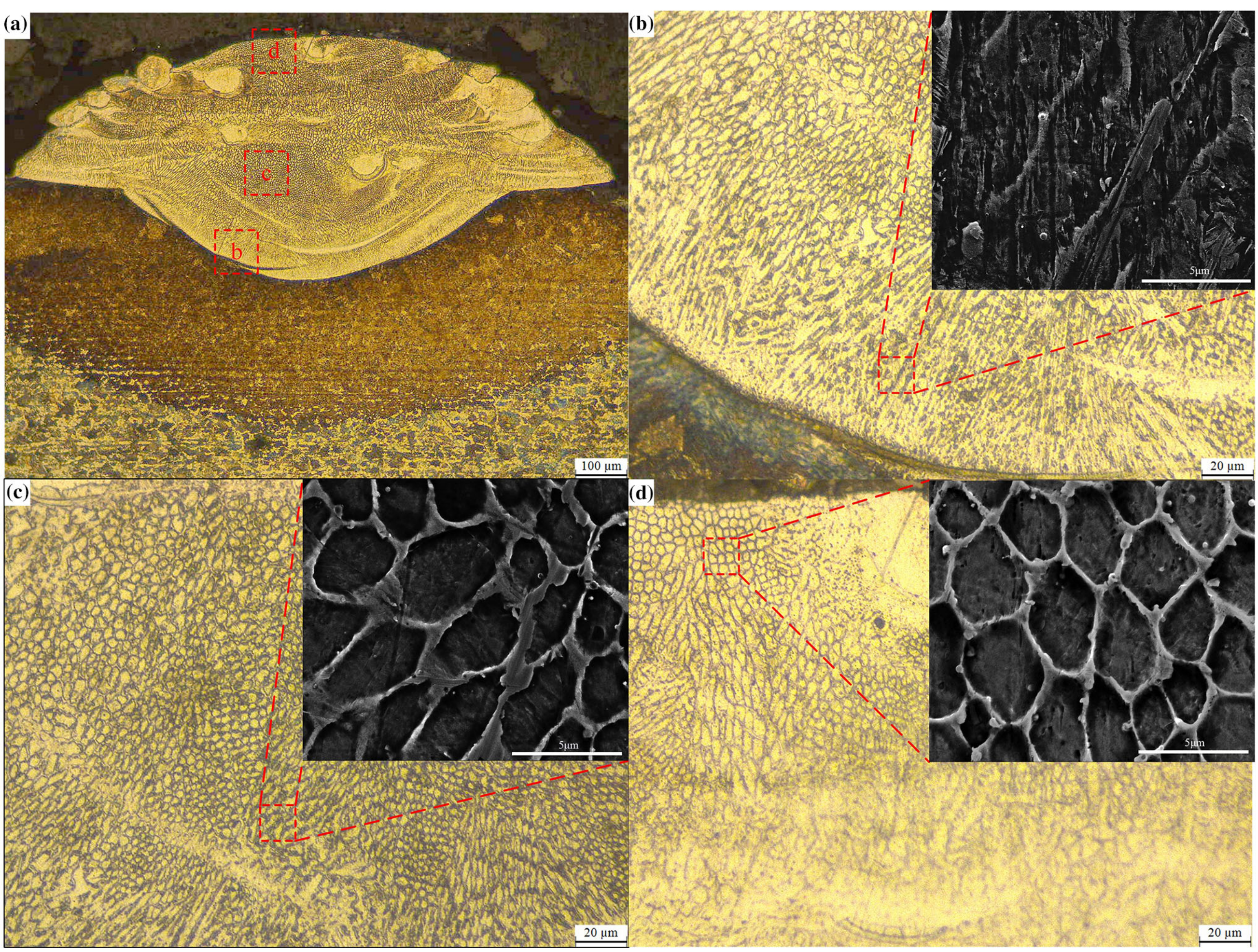


Fig. 7. Micrographs of different zones of the optimized cladding layer. (a) Overall cladding cross-section. (b) Bottom of the cladding layer. (c) Core of the cladding layer. (d) Top of the cladding layer

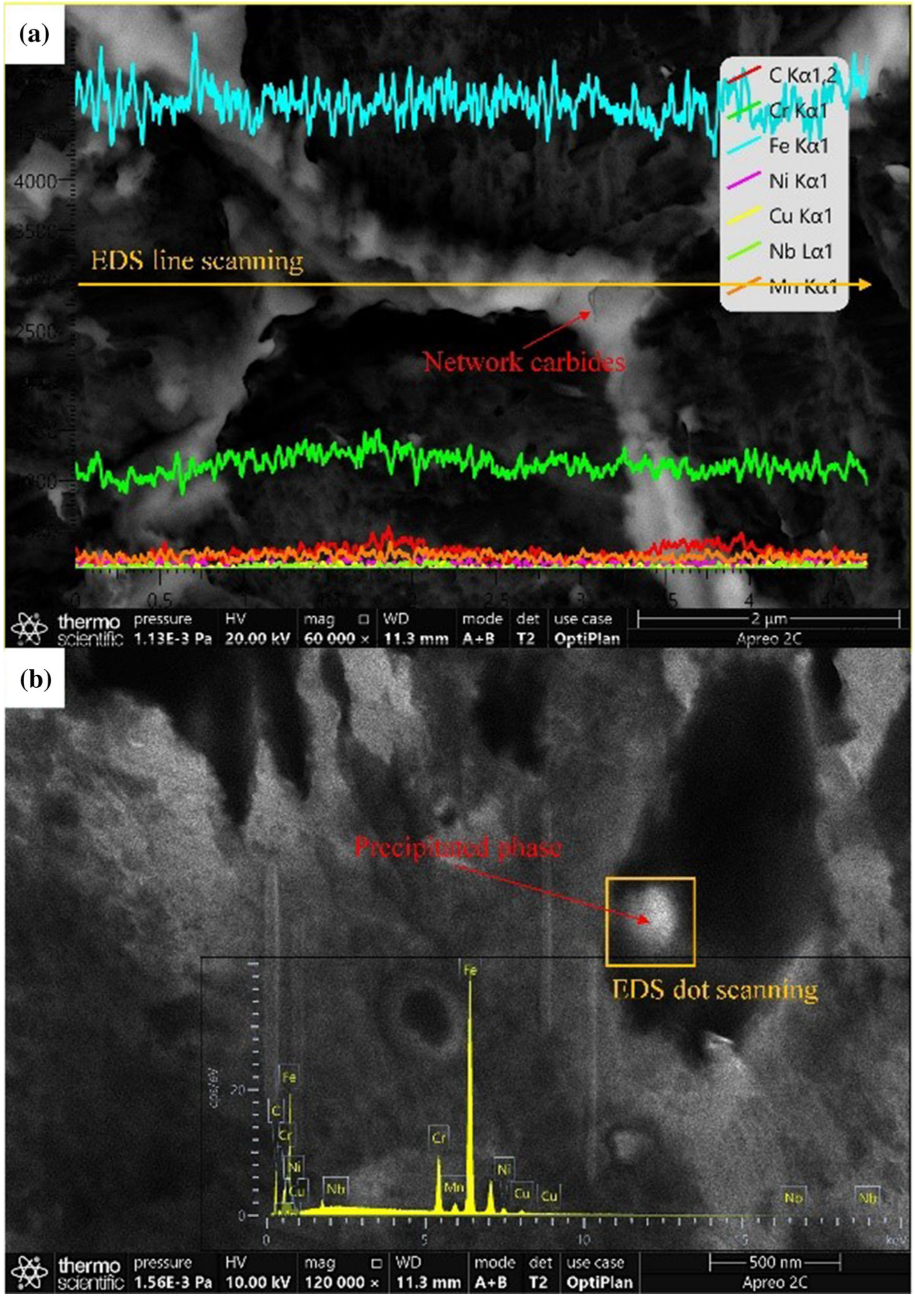


Fig. 8. The high magnification SEM and EDS results

2. Figure 7(c) shows the microstructure of the molten core, As the distance from the bonding interface region gradually increases, the solidification rate (R) increase and G/R value decreases. Thus the planar columnar crystals closed to the molten line play a certain hindrance to the heat diffusion from the middle of the molten pool. It helps the formation and growth of dendritic crystals, and further helps to form coarse dendritic crystals and cellular crystals along the direction of heat diffusion in the molten core. With the decrease of G/R, the strip carbide is changed into coarse network carbide and grown along the cooling direction. At the same time, fine spherical particles are gradually precipitated at the carbide boundary.
3. Due to that the upper part of the molten pool contacts with the air directly, rapid radiant heat dissipation forms a sharp temperature gradient, under which the solidification rate continues to increase. At this time, the cladding layer forms equiaxed crystals with fine grains and uniform size, which are shown in Fig. 7(d). The crystals in this part are smaller and denser than those in other parts, which play an important role in ensuring the hardness and wear resistance of the cladding layer (Ref 16, 33-35). At the same time, the mesh carbide boundary is shrunk and tended to be a hexahedral mesh with stable shape, fine spherical particles are dispersed in the network carbide boundary and the central Fe substrate.

The high magnification SEM photo of network carbides and fine spherical particles are shown in Fig. 8. In order to determine the elements and phases of these precipitates, the

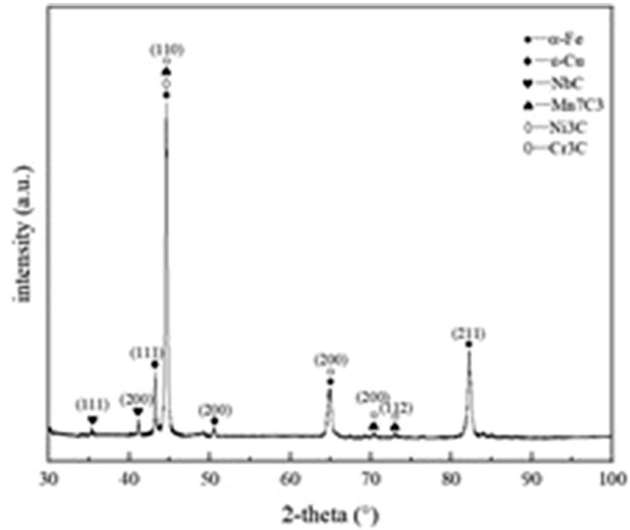


Fig. 9. XRD analysis for cladding layer

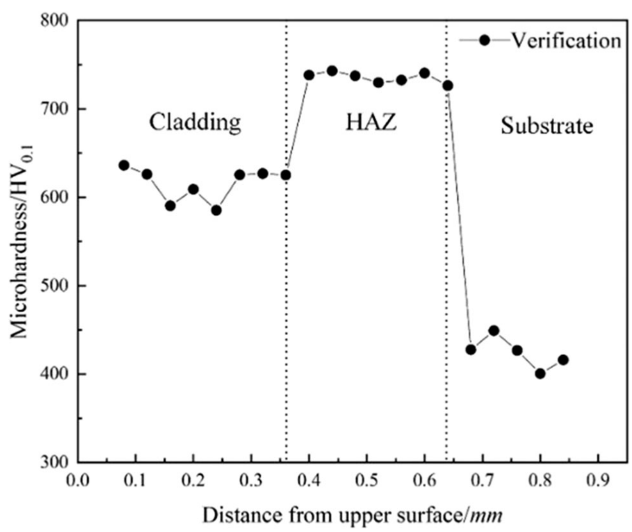


Fig. 10. The result of microhardness measurement

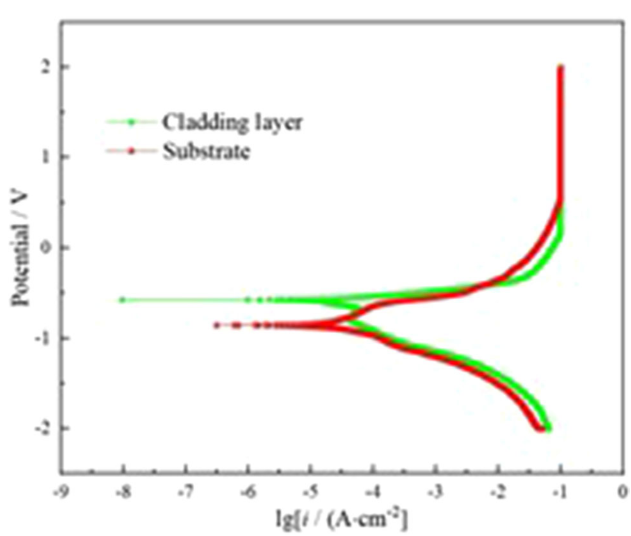


Fig. 11. Electrochemical corrosion polarization curves

energy dispersive spectrometer (EDS) is carried out on the boundary of network carbides and fine spherical particles respectively, and the results are shown in Fig. 8. Combined with the X-ray diffraction (XRD) analysis of the cladding layer in Fig. 9, carbides are mainly composed of large amounts of Cr in the form of M_3C, small amounts of Ni in the form of M_3C, with trace amounts of Mn and Nb in the form of M_7C_3 and MC respectively. At the same time, ε-Cu is precipitated out as fine spherical particles.

5.3 Microhardness Analysis

Pick a point every 0.04 mm from the upper surface of the cladding layer to measure its microhardness, three sets of parallel measurements are taken, and the average microhardness is finally taken. The measurement range is from Cladding layer—Heat Affected Zone (HAZ)—Substrate. The result is shown in Fig. 10. The analysis is shown as follows.

1. Compared with the hardness of the substrate, the hardness of the cladding layer is increased by about 1.5 times. The reason is that laser cladding has a high degree of undercooling under rapid melting and rapid solidification, insufficient time is given for crystal grains to grow and form fine grains to achieve the corresponding strengthen. At the same time, 15-5PH alloy powder and 45 steel are both Fe-based materials, but 15-5PH is rich in Cr, Ni, etc., which also has a certain influence on the hardness of the cladding layer.
2. The hardness of the HAZ is higher than that of the cladding layer. In the process of laser cladding, due to the high cooling rate of the molten pool, the grains in HAZ are refined significantly.

45 steel and 15-5PH are both Fe-based materials, and the composition of 15-5PH is rich in Cr and Ni elements. Under the action of high energy of laser beam, high hardness α-Fe martensite is generated in HAZ. On the one hand, other alloying elements such as Ni in 15-5PH may result in solid solution strengthening. On the other hand, some hardening phases, such as FeCr and $FeNi_3$, may be dispersing in HAZ.

Under the action of fine grain strengthening, solid solution strengthening, martensitic hardening and hardening phase dispersion strengthening, the hardness of HAZ is the highest. As the HAZ approaches the substrate, the dilution effect of the substrate becomes greater and greater, so the hardness tends to drop sharply.

5.4 Corrosion Analysis

Figure 11 shows the electrochemical corrosion polarization curves of the cladding layer and substrate under the optimal process parameters. The corrosion potential (E_{corr}) of the cladding layer is − 0.570V, and the corrosion current density (i_{corr}) is 2.699×10^{-5} A cm^{-2}. The E_{corr} and i_{corr} of the substrate are − 0.853V and 2.742×10^{-5} A cm^{-2}. The larger the E_{corr} and the smaller the i_{corr}, the higher the corrosion resistance.

The corrosion resistance of the 15-5PH cladding layer is slightly higher than that of the substrate. According to the line scan results of EDS is shown in Fig. 12, this is due to the rich corrosion resistant Cr element in the cladding layer. However, in the process of laser cladding, M_3C-type Cr carbide is precipitated and part of Cr element is consumed. which leads to

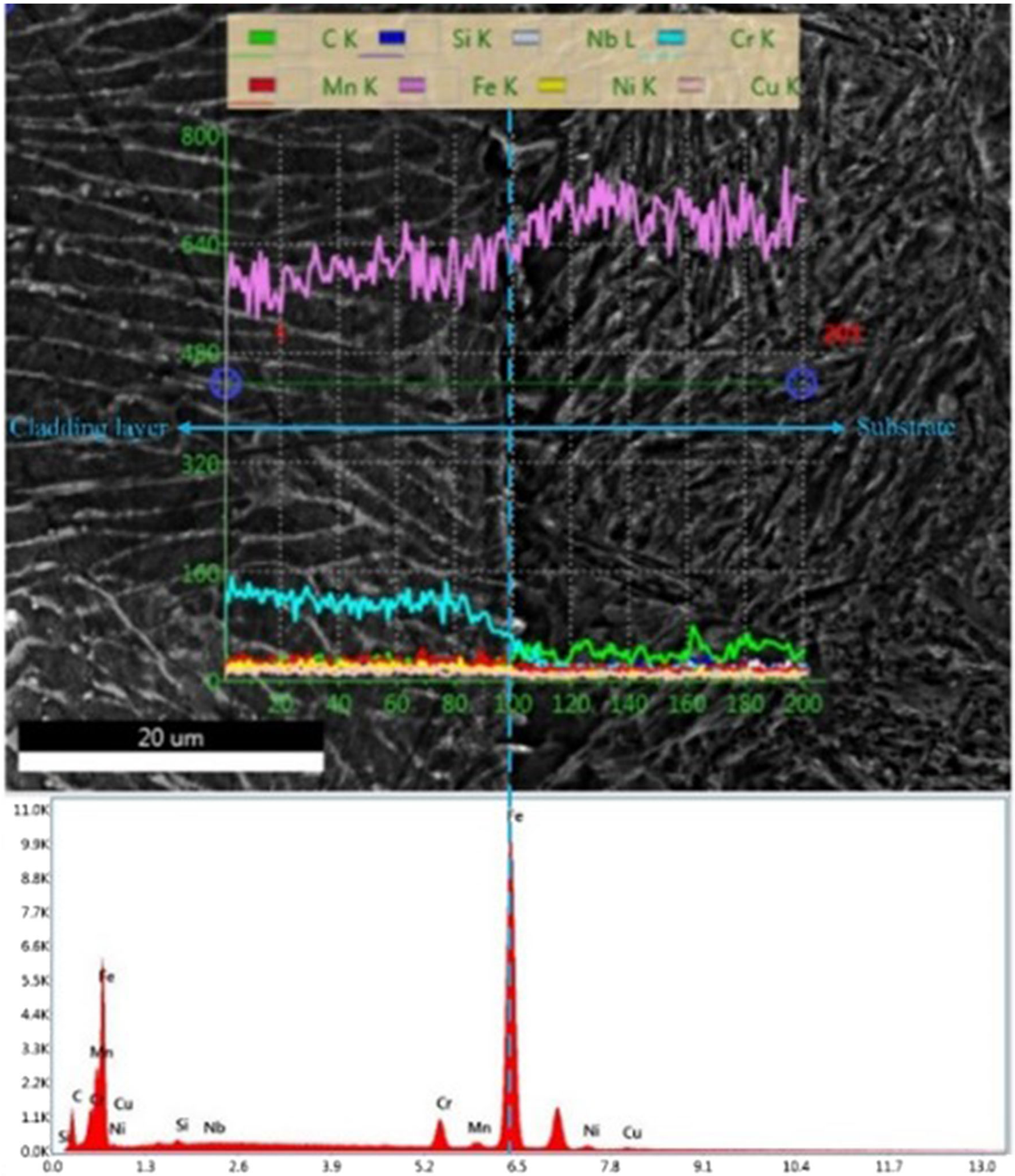


Fig. 12. EDS results of cladding layer and substrate

the decrease of Cr content which can form passivation film. Meanwhile, a small amount of Ni is diffused into the substrate, which slightly enhances the corrosion resistance of the substrate.

6. Conclusion

In response to the application of 15-5PH powder to the repair or remanufacturing of parts, a method is proposed to optimize the process parameters of laser cladding 15-5PH powder on the surface of 45# steel substrate. This method is based on Taguchi's method for experimental design, and combines AHP and EWM to determine the weights of the geometric quality characteristics of cladding layer. The comprehensive closeness index based on TOPSIS-GRA is introduced to analyze the influence of process parameters on the geometric quality characteristics of the cladding layer from Euclidean distance and situation. By ranking the experimental results, the best cladding layer with suitable geometric quality characteristics, dense microstructure, better microhardness and corrosion resistance could be obtained in a minimum possible time. Among the process parameters, the scanning speed (V) is the most significant parameter on the geometric quality characteristics of 15-5PH cladding layer. The performance of the cladding layer is improved mainly because of the network carbide and spherical ε-Cu precipitated during laser cladding.

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